Shell thickness in core-shell nanostructures

"What thickness range is used for MoS2 and ZnS shells in core-shell nanostructures?" *Generated by LeapSpace*

MoS_2 shells are used mainly in the ~1.3–20 nm range, while ZnS shells are commonly used in the ~3–46 nm range in the retrieved core-shell examples. The exact thickness depends on the core material and growth route, but the evidence consistently shows **thin few-layer MoS_2 shells** and **tens-of-nanometers ZnS shells** as practical regimes [145] [158] [159] [160] [161] [162] [163].

Thickness ranges seen in the evidence

Shell material	Thickness range reported	Typical "working" thickness in examples
MoS_2 shell	**1.3–3.6 nm** in Au@MoS_2 [158]; **4–16 nm** in MoO_2/MoS_2 nanoflakes [159]; **10–20 nm** in CdS@MoS_2 [160]; **~5–20 nm** in a MoS_2–ZnS hybrid [161]	Usually **few-layer to ~20 nm** [158] [159] [160] [161]
ZnS shell	**3–20 nm** in TiO_2/ZnS [145]; **6–35 nm** in ZnO/Ag_2S and related sulfide shells [163]; **7–46 nm** in ZnO–ZnS nanorods [162]; **~10 nm** in ZnO/ZnS [164]	Often **~10–20 nm**, but can extend to **tens of nm** [145] [162] [163] [164]

Practical takeaway

- **MoS_2** is usually kept **thin** when it functions as an outer active shell, often at **few-layer thicknesses** or up to **~20 nm** [158] [159] [160].
- **ZnS** shells are generally **thicker than MoS_2 shells** in the examples here, commonly **around 10 nm** and sometimes up to **~46 nm** depending on the system [145] [162] [163] [164].
- In both cases, the evidence suggests that **intermediate thicknesses** are often the useful design point rather than maximally thick shells [145] [162] [163].

Bottom line: a reasonable working target from these precedents is **MoS_2 ≈ 1–20 nm** and **ZnS ≈ 3–46 nm**, with **MoS_2 usually optimized thinner than ZnS** in core-shell nanostructures [145] [158] [159] [160] [163].

Confidence level: High — multiple independent core-shell studies report thicknesses in these same bands, though the best value still depends on the specific core, shell sequence, and target function.

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